10. Applying Performance Optimization Techniques (Indexing, Subforms, and Query Tuning) in Zoho Creator

Aim

To study and apply performance optimization techniques such as field indexing, efficient subform structuring, and Deluge query tuning in the **Student Admission Enquiry Management** application developed using Zoho Creator, with the goal of improving enquiry search, form submission, and report responsiveness.

Overview

The Student Admission Enquiry Management application stores student details, course preferences, academic information, sibling records, and admission statuses. As the number of enquiries increases, careful field configuration and efficient data retrieval help reduce unnecessary operations and maintain responsive reports.

1. Field Indexing

Concept: Indexing frequently searched fields can improve database lookup performance. In this project, candidate fields include Admission Enquiry ID and Email Address, subject to the indexing options available in Zoho Creator.

Benefit: Faster record identification and targeted filtering when processing many enquiries. A uniqueness constraint by itself should not be treated as proof of indexing.

2. Efficient Subform Usage

Concept: The existing **Sibling Details** subform stores Sibling Name, Current Class, and School Name as related rows within the Admission_Enquiry_Form.

Benefit: Related sibling information remains organized within one admission record, avoiding redundant separate forms while minimizing unnecessary fields.

3. Deluge Query Tuning and Best Practices

Concept: Optimized Deluge workflows use specific search criteria, avoid repeatedly retrieving all admission records, and minimize database operations inside loops.

Benefit: Reduced processing overhead for workflows such as enquiry status tracking, counselling notifications, and validation. Actual improvements require performance tests.

Procedure

Step 1: Identify Fields for Efficient Searching

• Open Zoho Creator and select **Student Admission Enquiry Management**.

• Open the **Admission_Enquiry_Form** in the form designer.

• Review the properties of searchable fields such as Admission Enquiry ID and Email Address.

• Where supported, configure suitable unique or index options and save; do not apply uniqueness to fields where duplicates are valid.

• Verify that search and filtering still return the correct student enquiry records.

Step 2: Optimize the Sibling Details Subform

• Open the existing **Sibling Details** subform.

• Keep the essential columns: Sibling Name, Current Class, and School Name.

• Avoid adding redundant or unnecessary subform fields and expensive on-change scripts.

• Preview the form and verify that adding a sibling row works correctly.

Step 3: Improve Deluge Data Retrieval

• Open **Workflow** and inspect relevant form workflows or Deluge functions.

• Replace unnecessarily broad record retrieval with precise criteria where applicable.

• Avoid fetching the same enquiry records repeatedly inside for-each loops.

• Where supported, limit retrieved rows or process larger sets in controlled batches.

• Save and test each changed script against sample records.

Illustrative query patterns (adapt to actual field link names):

```
Broad: enquiries = Admission_Enquiry_Form[ID != 0];

Targeted: enquiries = Admission_Enquiry_Form[Enquiry_Status == "Enquiry Received"];
```

This is an illustrative Deluge filter, not a tested script for the current project. Confirm the exact link name of Enquiry_Status before use.

Step 4: Test and Monitor Performance

• Record baseline loading time for the All_Enquiries_List report and submission time for Admission_Enquiry_Form.

• Use available execution logs and debugger tools to inspect workflow execution.

• Repeat the same tests after modifications, using comparable records and conditions.

• Compare response times, check for errors, and document observations without inventing improvement percentages.

Application-Specific Example

When an Admission Cell user searches for a student or filters by Enquiry Status, a targeted lookup can reduce the need to scan unrelated records. The Sibling Details subform keeps repeated sibling information together, while tuned Deluge workflows can reduce unnecessary operations during validation and notification.

Result

Performance optimization methods involving field indexing, sibling subform design, and targeted Deluge queries were studied and documented for the Student Admission Enquiry Management application. These methods provide a practical approach for improving scalability and responsiveness as admission enquiry volumes increase. **Measured speed improvements and successful indexing configuration remain to be verified through testing.**